Course Code: CSE233/CSE306	Course Title: Cloud Computing Type of Course: Theory	L- T-P-C	3-0	0	3
Version No.	1				
Course Pre-requisites	Basics of Distributed Computing, Service Oriented Architecture				
Anti-requisites	nil				
Course Description	This Course is designed to impart the knowledge of Cloud Computing as a new computing paradigm. The course explores various Cloud Computing terminology, principles and applications. The course also demonstrates the different views of the Cloud Computing such as theoretical, technical and commercial aspects.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Cloud Computing and attain Employability through Participative Learning techniques.				
Course Out Comes	On successful completion of the course the students shall be able to: Describe fundamentals of cloud computing, virtualization and cloud computing services. Explain security and standards in cloud computing. Discuss Cloud mechanisms to optimize the QoS parameters. Develop applications using Cloud services and VM instances.				
Course Content:					

Module 1	10 Sessions
Introduction to Cloud Cloud Computing at a Glance, Historical Developments, Building Cloud Computing Environments, Computing Platforms and Technologies, Technology Examples, Cloud Computing Architecture, IaaS, PaaS, SaaS, Types of Clouds, Economics of Cloud	
Module 2	10 Sessions
Virtualization Techniques Basics of Virtualization - Types of Virtualizations, Taxonomy of Virtualization Techniques, Implementation Levels of Virtualization.	
Module 3	09 Sessions
Cloud QoS and Management Cloud Infrastructure Mechanisms, SLAs, Specialized Cloud Mechanisms, Cloud Management Mechanisms, Cloud Security Mechanisms.	

Module 4	09 Sessions
Cloud Platforms, Advances in cloud: introduction to Amazon Web Services: Introduction to Google App Engine, Introduction to Microsoft Azure. Media Clouds - Security Clouds - Computing Clouds - Mobile Clouds – Federated Clouds – Hybrid Cloud	
Text Book John Rittinghouse and James Ransome, “Cloud Computing, Implementation, Management and Security”, CRC Press. Rajkumar Buyya, Christian Vecchiola, and Thamarai Selvi, “Mastering Cloud Computing”, McGraw Hill Education.	
References David E.Y. Sarna, “Implementing and Developing Cloud Applications”, CRC Press. Anthony T Velte, Toby J Velte, Robert Elsenpeter, “Cloud Computing: A Practical Approach”, Tata McGraw-Hill. Web resources: https://puniversity.informaticsglobal.com:2229/login.aspx	
Topics relevant to development of “Skill Development Aws, Azure, APIs, Aneka Cloud Platform, EC2, Installation of VM Workstation, Infrastructure Security Challenges for Skill Development through Participative Learning techniques. This is attained through assessment component mentioned in course handout.	